FALCON AUTOTECH

FALCON's Response to Delhivery's RFQ for Hyderabad Warehouse Automation for Parcel Sortation

April 1, 2023

www.falconautotech.com

Kind Attention

Mr Sunny Raja
Mr Himanshu Sekhar
Delhivery

Offer Ref: F23-00173-02; Date 01.04.2023.
Subject – Techno-commercial Offer for the Cross-Belt Sorter_12k for Hyderabad Site

Dear Team,

We are pleased to submit our Techno-commercial offer in response to your RFQ.

As you will note, we have studied your requirement in great depth and, along with the information collected during calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposal and to re-enforce our commitment to be your partner in this important project.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Sortation Technologies and references.

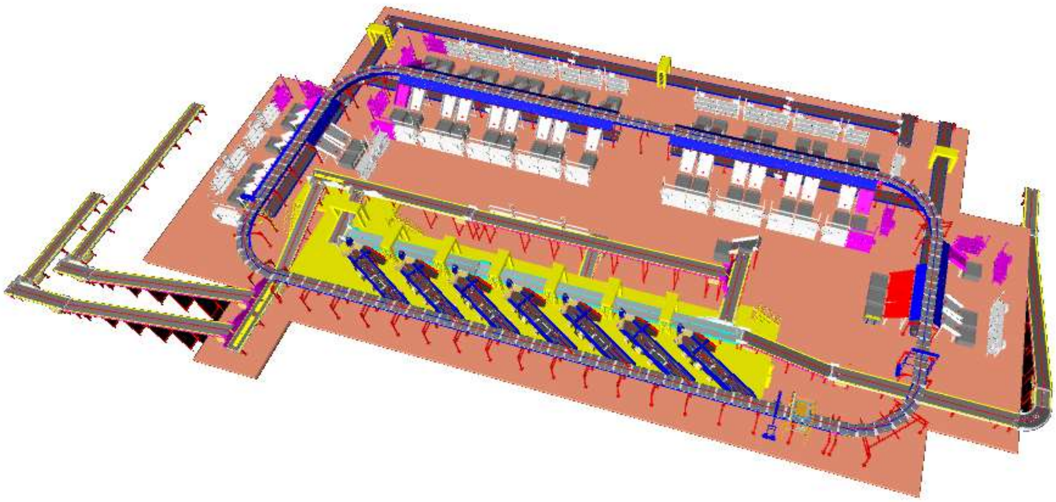
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through your award process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

Best Regards,

Adib Hasim

Manager – Sales

Response to RFP for Cross Belt Sorter_12K_Delhivery_Hyderabad



Proposal Reference - F23-00173-02

Falcon Autotech Private Limited
Plot No 308 & 309
Ecotech-1 Extension, Greater Noida
Uttar Pradesh- 201308
India.

Contact – Adib Hasim
Sales Manager
Mob - +91 9711619998
Adib.hasim@falconautoonine.com

Table of Contents

1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Parcels Par Hour
3	ARB	Activated Roller Belts
4	DBO	Damaged Barcode
5	VDS	Volume Distribution System
6	ICR	Intelligent Character Recognition
7	MEZZ	Mezzanine
8	LIM	Linear Induction Motors
9	ECDS	Empty Carrier Detection System
10	AC	Alternating Current
11	DC	Direct Current
12	PLC	Programmable Logic Controller
13	IT	Information Technology
14	BOQ	Bill Of Quantity
15	I/O	Input/ Output
16	PDP	Power Distribution Panel
17	PC	Personal Computer
18	UPS	Uninterrupted Power Supply
19	CBS	Cross Belt Sorter
20	DNF	Data Not Found
21	LGM	Logic Mismatch
22	RTVC	Real Time Video Coding

2. Company Profile

THE COMPANY

OUR VISION

Create a Global impact on the society by facilitating World's leading Organizations run their Supply Chains efficiently by leveraging technology enabled products and Solutions and, in the process, establish Falcon Autotech as one of the World's most respected and trusted brand in Intra-logistics Warehouse Automation Space.

ABOUT US

The team started out in 2004 solving Special Purpose Automation problems for clients and later pivoted to building Standard Technology Stack spanning across hardware, firmware and Software to tackle bigger Supply Chain Problems.

Over the last few years, Organization has made rapid strides and has carved out a niche in some of the world's most cutting edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

TEAM

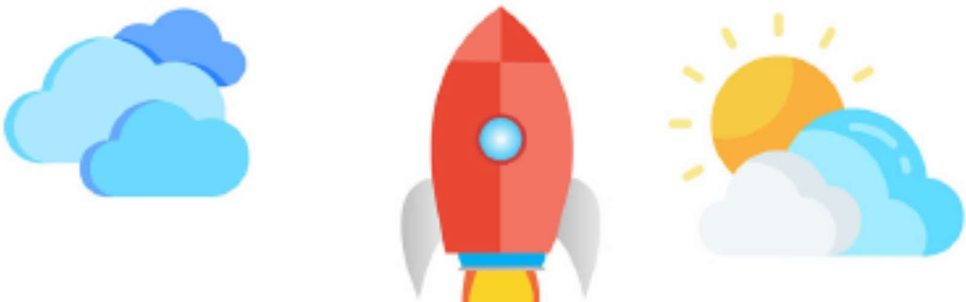
Organizations are built and defined by the people they are able to attract, cultivate and retain and the talent management philosophy at Falcon Autotech could not agree more.

At, Falcon Autotech we take a lot of pride in being the preferred career choice for some of the finest talent in the country across multiple domains such as:

- Supply Chain Solution Designing and Consultation
- Hardware Design
- Electrical Design
- Electronics Design
- Software Development
- Manufacturing

Giving us an unprecedented ability harness true potential generated by culmination of Supply Chain Know How, Hardware and Software under one roof.

COMPANY HISTORY



2021

1. Company expands its manufacturing footprint by 4X.
2. Team size continues to grow, with 400+ on-roll employees.
3. Orders booked for some of the largest projects being executed in India.

2018

1. Company sets up its first Sorter in Europe.
2. Company announces representation in North America.
3. Company ranked amongst the top 20 global players in the sorting automation market.

2017

1. Company bags some of the biggest Warehouse Automation projects in India.
2. Company launches Cross Belt Sorter and Pop-Up Sorter.
3. Company expands its representation in UK, South East Asia, Middle East, South Africa, Eastern Europe and China through partners.

2016

Company starts exporting to 12 countries in APAC, GCC, USA and China.

2015

1. Company launches a series of products: Tilt Tray Sorter, Linear Arm Sorter, Pick/Put to Light System.
2. Company bags several high-profile automation projects from Blue Dart, DHL, Delhivery, Aramex, Flipkart etc.

2014

1. Company moves to a new manufacturing unit in Greater Noida.
2. Company bags its first order of Sortation System and Put to Light System from a leading Express Logistics player in India.

2013

Company launches and deploys its first product, 'Dimension and Weight Scanning System' at Jabong.com (now a subsidiary of Walmart FlipKart group).

2012

A standard Products division is launched under the brand name of **Falcon Autotech.**

2010

Team grows to 45 people and the Company starts working with multiple clients in Auto, Pharma and Consumer Electronics space.

2005

Company bags its first order worth INR 1 Million from an automotive component manufacturer.

2004

Starts as a customized machine builder (Special Purpose Machine) with a team of 2, under the brand name of **Futuristic Robotics.**

KEY STATISTICS

04
Product Lines

400+
On-Roll Employees

215+
Engineering Team

~200000 Sq Ft
Manufacturing Space

20+
Tier A Clients

15+
Countries with Live Installations

1700+
Total Installed Systems Globally

35
Representations in 4 Continents and 35 Countries

1Million+
Sorts Per Hour of combined Sorting Capacity installed Globally

OUR PRODUCT PORTFOLIO

01 SORTATION SOLUTIONS


Cross Belt Sorter


Tilt Tray Sorter


Arm Sorter


Swivel Wheel Sorter


Sweep Sorter


Pusher Sorter


Pop Up Sorter

02 PICK/PUT TO LIGHT SYSTEMS


Packet Sortation


Multi-Line Order Consolidation


Order Picking


Store Order Fulfillment


Spare Parts Distribution


Kitting


Multi Order Pick Cart

03 DIMENSION & WEIGHT SCANNING SYSTEMS

STATIC DIMENSION & WEIGHT SCANNING SYSTEMS


Cubizon-R


Cubizon-R THRU


Cubizon-R Cross


Cubizon-R ECO

DYNAMIC CONVEYOR BASED SYSTEMS

OUR 8 MODELS

OUR PRODUCT PORTFOLIO

04 CONVEYOR AUTOMATION SOLUTIONS

BELT CONVEYORS

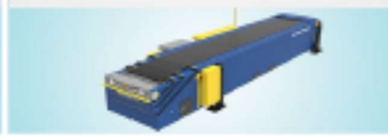
- **PVC BELT**
  - Straight Conveyor
  - Spacing Conveyor
  - Accumulation Conveyor
  - Strip Belt Merge- 30/45/60
  - Solid Belt Merge- 45
  - Incline Belt Conveyor
- **MODULAR BELT**
  - Straight Conveyor
  - 90 Degree Belt Curve
  - 180 Degree Belt Curve

ROLLER CONVEYORS

- Straight Conveyor
- Gravity Roller Conveyor
- MDR Accumulation
- Angled Roller Conveyor
- Curved Conveyor
- Spur Merge Conveyor
- Swivel Wheel Diverter
- Pop Up Belt Diverter
- Pusher Unit

SPECIAL APPLICATION SOLUTIONS

- Aligning Conveyor
- One to Two Split Conveyor
- Two to One Merge Module
- Bulk Distribution-ARB
- Bulk Distribution-Telescopic
- Bulk Distribution-Electric Arm
- Telescopic Belt Conveyor



05 SOFTWARE SUITE

FALCON'S AUTOMATION CONTROL SUITE

FACtS DASHBOARD

ControlIT — WAREHOUSE CONTROL SYSTEM

- **CUBIZONE™** — 'Cubizone' is a fully integrated software bundled with our Cubizon Series for dimensioning & weighing.
- **MOVEIT** — MOVEIT is the core software for conveyor modules to execute the functional process of scanning, dimensioning etc.
- **SORTIT** — SORTIT is the sorting engine to execute the functional process of sorting, dimensioning etc.
- **LIGHTIT** — LIGHTIT is the core software that runs Falcon's Pick/Put to light systems.

OPERATIONS ANALYTICS

KEY FEATURES OF OPERATIONAL ANALYTICS

AUTOMATION MANAGEMENT SYSTEM

- **PREVENTIVE MAINTENANCE CHECKLISTS, ISSUE LOGGING & TROUBLESHOOTING** — Mobile App based Preventive Maintenance routines & Issue Logging.
- **PREDICTIVE MAINTENANCE USING MACHINE LEARNING** — An always-on data-crunching tool, that analyses component level performance trends & alerts user on anomaly detection (before failure).
- **LEARNING MANAGEMENT SYSTEM** — An all-in-one Training Portal encompassing User Training, PM Training, Circuit Diagrams, Assembly Video Tutorials and much more.
- **SPARES AND WARRANTIES MANAGEMENT** — A single window to manage and control multi-site Spares, Warranties, AMCs, Service visits etc.

ADD-ON MODULES

Real Time Video Coding

No-Read Barcodes have always been a challenge in Sorter Applications globally. With Falcon's RTVC feature, we can now easily decode such challenging parcels in real time while sitting remotely.

Deep Learning Image Classification

DLIC is an artificial intelligence function that imitates the working of the human brain. It can identify objects such as packets on the carrier and classify them in different categories.

Cloud Sort App/Virtual PTL

It's an android based app used to perform secondary sorting in light assisted put-away operations. It can work as a stand-alone setup or can be integrated with any sorter.

INDUSTRY SOLUTIONS

As a leading player in Intra Logistics Automation Space, Falcon Autotech continuously strives to improve the operational efficiency and accuracy for its clients through its domain knowledge consultancy and experience in addition to its wide range of Products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in Select Industries that lie in the Forte of Falcon Autotech and deliver world class projects in return.

FOCUS INDUSTRIES

E-COMMERCE RETAIL

COURIER EXPRESS POSTAL

FASHION DISTRIBUTION

FFP (FOOD,FMCG,PHARMA) DISTRIBUTION

SPARE PARTS DISTRIBUTION

GROCERY DISTRIBUTION

FOCUS APPLICATIONS FOR FALCON AUTOTECH

- Piece Picking and Handling
- Case Picking and Handling
- Parcel Handling

SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.

MEDIA COVERAGE

NO.1 LOGISTICS AUTOMATION COMPANY IN INDIA*

Falcon Autotech has achieved high recognition in the Intra-Logistics Automation world, "Global Logistics Automation Market" report published by Data Bridge Market Research has featured Falcon Autotech among top 20 in global list of Sortation Systems manufacturers. CNBC Awaaz, Awaaz Entrepreneur featured Falcon Autotech as fastest growing Logistics Automation company in India, Besides being featured in many important Daily Newspapers, we have also won the "Best Warehouse Automation Company" of the Year award at Express Logistics Supply Chain Conclave held in Mumbai in the Month of September 2015.

THE HINDU BusinessLine

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Robotics powers festival sale impact on productivity for e-commerce brands

JESSU JOHN · PRINT · T+

October 16, 2016: Between Amazon's 'Great Indian Festival', Snapdeal's 'Unbox Diwali' and Flipkart's 'Big Billion Day' sales, all eyes were on deals and wins earlier this month. However, what's common for most brands executing large impact sales seasons is the need for operational efficiency. Falcon Autotech, an industrial automation solution provider in India, enables efficient fulfilment in the face of a surge of orders experienced by many online retailers during these high demand periods.

Vineet Baid, CEO, Falcon Autotech, explains, "It's the warehouses or fulfilment centres that see 2 to 2.5 times the regular volumes for large festive sale periods."

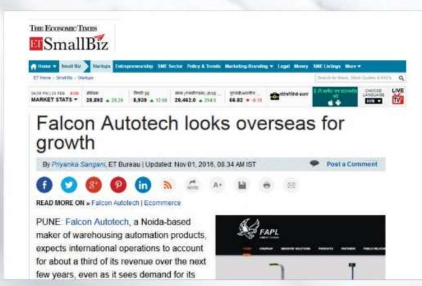
EFFICIENCY UNPLUGGED

THE FINANCIAL EXPRESS

How automation firm Falcon helping clients deliver products to workers

Industrial automation firm Falcon is helping its clients use robotic carts to pick, place and deliver products from specific locations to workers

Ever wondered how an e-commerce company fulfills thousands of orders everyday—picking up the product from the warehouse nearest the delivery address, packing it properly, labelling it with the right barcode and mandatory receipts, and then ensuring that it reaches the customer in minimal time? While an army of men and machines work round the clock to make it happen,



THE ECONOMIC TIMES ET SmallBiz

Falcon Autotech looks overseas for growth

By Priyanka Sangani, ET Bureau | Updated: Nov 01, 2016, 08.34 AM IST · Post a Comment

READ MORE ON » Falcon Autotech | Ecommerce

PUNE: Falcon Autotech, a Noida-based maker of warehousing automation products, expects international operations to account for about a third of its revenue over the next few years, even as it sees demand for its

THE ASIAN AGE

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Hidden technologies changing our shopping experience

Ecommerce sites finally solve the conundrum of sustainable growth and profitability

AGE CORRESPONDENT

Ecommerce sites have been able to gain traction thus far, by indulging shoppers in cash-backs and discounts. However, the same had a cobra effect on the economy, doing more harm than good, raising doubts on the authenticity of these sites.

Ecommerce sites of today seem to finally solve the conundrum of sustainable growth and profitability, by extending the most intuitive and simplified experience. Since technology has always come to the rescue of economy, let us delve deeper and get acquainted with tech-startups assisting ecommerce sites extend a customer experience so simplified, it's borderline

matching system simplifying product search via uploading images, also offering most suitable

ecommerce sites sprouting up, consumers today have to battle with decision fatigue, often result-

Tackling warehousing concerns via automation

It is pivotal that the right product reaches the right doorstep, since every delivery is unquestionably paramount.

Understanding the warehousing activities, Falcon Autotech ensures the most efficient product delivery.

Introducing deep Automation Roots, Falcon is one amongst the leading solution providers in India, automating the warehousing tasks in ecommerce retail, warehousing, logistics, including courier and parcel, food and FMCG, pharmaceutical and automobiles.

Organising the chaotic

DECCAN Chronicle

TECHNOLOGY, IN OTHER NEWS

These tech oriented companies are making it smooth for industries to run

DECCAN CHRONICLE

Published Oct 27, 2016, 5:05 pm IST Updated Oct 27, 2016, 5:05 pm IST

Industries still need to cross a lot of hurdles. Technology is playing role in every beat and is innovating to resolve the challenges. Here are few organizations that have made it better for industries in India. Not only have they saved time but are also saving a chunk of money for different industries.

Aaveg manages the entire activity chain including Employee transportation, Leasing to business rental and advising, managing and procuring the overall

PRODUCT CARGOTALK 37

Cutting edge tech for dimensioning & weighing

The highly accurate and versatile Cubizon-R THRU is equipped with roller bed base, has robust operations along with guaranteed zero-maintenance.

CT Bureau

Falcon Autotech has announced the launch of Cubizon-R THRU in the country. The product with cutting-edge technology is specifically designed for cargo applications that require dimensioning and weighing of heavy bulky packets. Cubizon-R THRU is also equipped with roller bed base to provide it an unmatched functionality and gives its users the ability to manoeuvre typical freight items which are hard to manoeuvre otherwise.

Cubizon-R THRU is equipped with high precision laser for ultra-high accuracy and has integrated slider flaps that enable accurate measurement of irregular products. It can accurately measure regular and irregular products with minimum product dimension 50X50X50 mm to 1200X1000X1000 mm and can

Cubizon-R THRU helps a company with warehouse storage space planning and shipment freight costs

functionality and unmatched user experience, the power-packed Cubizon-R THRU is an end-to-end solution for all of your weighing and dimensioning needs for the category." Cubizon-R THRU helps a company with warehouse storage space planning and optimisation, shipment freight cost calculation and reconciliation, packaging size recommendation and optimisation, and size recommendation for modern day delivery models such as smart lockers.

OVER 100 YEARS

100 YEARS OF GIVING

*Source: Report by Data Bridge Market Research

CLIENT TESTIMONIALS

“ Working with Falcon Autotech has been a very delightful experience for us, BLUE DART being a leader in the express logistics business, we expect the same kind of responsiveness, reliability, commitment & passion from our partners and Falcon has always delivered to their promise on all the solutions such as the Auto Sorter, Weight Dimension Labelling machine, Parcel Locker etc. All the technology interventions have been business critical for us.

Anil Khanna, Managing Director
DHL BLUE DART

“ Falcon is an organisation that has shown great agility and has helped us improve our service capabilities. Their sortation systems have helped us deal effectively with our growing volumes. We have found the Falcon team quite responsive to feedback and quick to incorporate changes into their systems that have helped build better, customized solutions for Delhivery.

Sahil Barua, CEO
DELHIVERY

“ Falcon Autotech has been efficient and proactive, particularly in meeting deadlines and their technical advancements. Falcon Autotech strike just the right balance between being a flexible, easy to deal with firm yet having all the key strengths and attributes. Every member of the team is honest, reliable, responsive, personable and very knowledgeable.
I wish them good luck for all the future endeavours.

Sanjeev Saxena, Director
ECOM EXPRESS

“ I believe you guys earned it with your hard work and dedication. I believe the products you have designed demonstrate innovation and lateral thinking. Also the management has a flexible and the customer centric approach. It has been a delight to work with you and the team.

Amitava Saha
CEO, firstcry.com

3. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers** globally. Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolio in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **6 Million Parcels parday** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.

Born2Invest
Save It. Invest It. Prosper.

Incas-SSI Schaefer merger creates leading player in Italy's intralogistics market

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expected to have a higher CAGR. This is due to factors associated with the growing need to deliver goods in a timely manner and to reduce transportation costs.

Moreover, automotive will be the leading segment of the industry from 2018 to 2023, with an increasing demand for automated production systems and flexible logistics systems used in manufacturing and supply of vehicle fleets.

The MarkestandMarkets research report has identified SSI Schaefer as one of the key players in the logistics automation market worldwide during the forecast period, along with Dematic Corporation (US), Murata Machinery, Ltd. (Japan), Jungheinrich AG (Germany), and Falcon Autotech (India), among others.

This year's largest market for logistics automation is estimated to be North America, according to the research report. Regardless, Incas and SSI Schaefer's merger could help them compete in their main market.

(Featured image by DepositPhotos)

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SHARES

Not secure | worldindustryinsights.com/30269/global-parcel-sortation-system-market-analysis-size-share-growth-trends-and-forecast-2019-2026-siemens-vanderlande-beumer-group

Global Parcel Sortation System Market Analysis, Size, Share, Growth, Trends, and Forecast 2019-2026: Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives

BY ANN.CASTRO ON AUGUST 12, 2019

The global "Parcel Sortation System market" research report is a pervasive research report that touches the most vital parts of the Parcel Sortation System market that is necessary to be grasped by a professional or even a layman. The market research report enlightens one regarding few of the important aspects such as an overview of the Parcel Sortation System product, the growth factors enhancing or hampering its development, application in the various fields, major dominating companies, genuine facts, economic situation, and geographical analysis. The research report endows the data concerning the dynamics that propel the growth as well as the supply and demand chain of the product on a global basis.

Click here to access the report:: www.reportsbuzz.com/request-for-sample.html?repid=50822

The international and global stand of the Parcel Sortation System market is also briefly mentioned in the research report based on the performed statistical and thorough market analysis. The information mentioned in the research report gives a qualitative and quantitative view of the overall market. The statistical analysis of the market helps analyze the supply, demand, production, maintenance, and storage costs of the product. The data regarding some of the prevalent players Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives, Dematic, Interroll, Muratec, Invata Intralogisitcs, Viastore, Okura Yusoki, GBI Intralogistics, Equinox Mhe, Pitney Bowes, Falcon Autotech, OCM SRL, Solystic, Bowe Systec, Intralox are also detailed in the current case study.

4. Reference projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Piece Picking and Handling, Case Picking and Handling & Parcel Handling
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

4.1 Project 1- (Ecommerce Client)

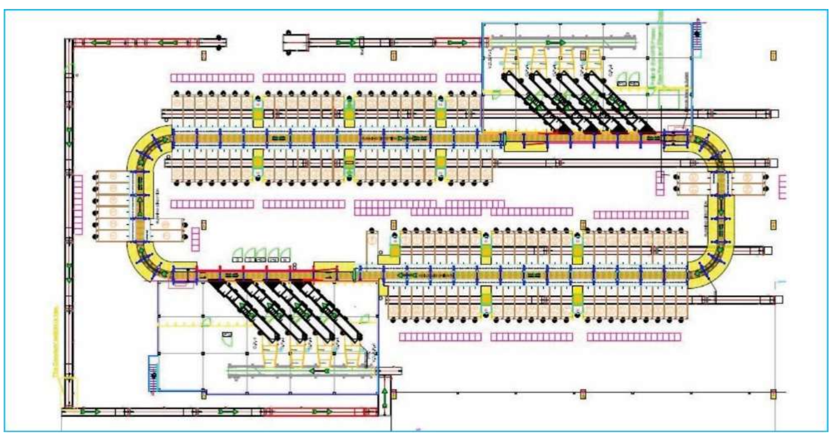
Use Case - Destination Sorting of Packed Parcels

Solution Specifications –

- Throughput: 16,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



Layout & Pictures_ Project 1

4.2 Project 2- (Client – Courier Express Parcel)

Use Case - Courier Sorting of Parcels

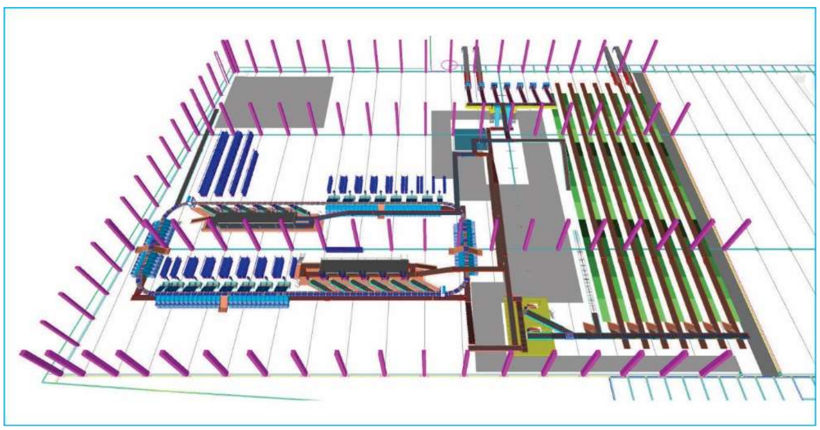
Solution Specifications –

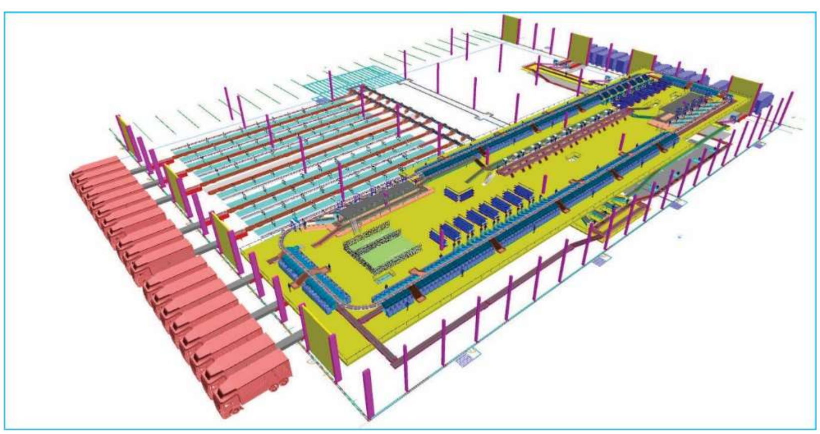
- Throughput: 24,000 PPH
- End Destinations: 394 Direct, 30 Secondary, 630 PTLs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Pre-Sorting
- Telescopic Belt Conveyors
- Bag Sorters
- PTL System

Layout -





Layout & Pictures_ Project 2

4.3 Project 3- (Client – E-Commerce)

Use Case – Destination Sorting of Packed Parcels

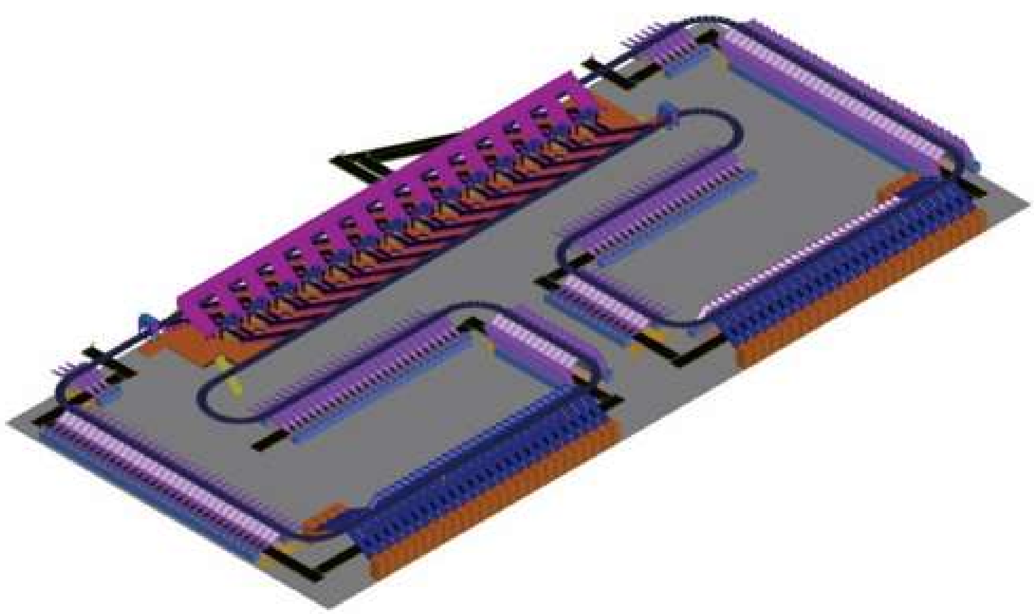
Solution Specifications –

- Throughput: 26,000 PPH
- End Destinations: 410
- Building Size: 500,000 Sq Ft

Key Technology Modules –

- Bulk Flow Sorting
- Automatic Pre-Sorting
- Loop Cross Belt Sorter with cell size of 900 x 500 mm
- Bag Conveyors

Layout -



Layout _ Project 3

4.4 Project 4- (Client – Indian Retailer)

Use Case – Store wise Sorting of Apparels

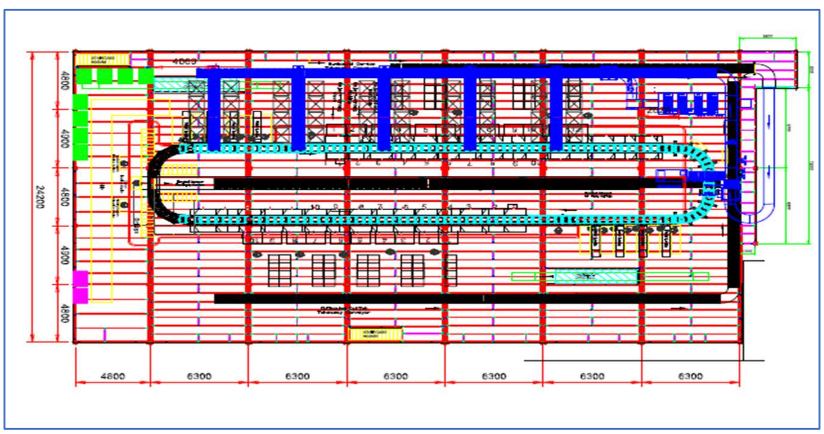
Solution Specifications –

- Throughput: 20,000 PPH
- End Destinations: 1750 using PTLs
- Building Size: 700,000 Sq. Ft

Key Technology Modules –

- Loop Cross Belt Sorter with a cell size of 600mm x 330 mm
- Belt Conveyors
- Case Sorters
- Telescopic Belt Conveyors
- Packing Machines
- Vertical Lifters

Layout and Site Pictures –



Layout & Pictures_ Project 4

5. Handled Parcel Spectrum

5.1 Parcel size loadable on the sorter

Max Length	mm	500
Max Width	mm	500
Max Height	mm	500
Max Weight	Kg	10
Min length	mm	100
Min Width	mm	50
Min Height	mm	5
Min Weight	Kg	0.02
Dimension Accuracy	mm	(+-10)
Weight Accuracy	Kg	(+-0.06)

Note: Min measurable size by dimension scanner will be 100x50x20 (mm).

5.2 Parcels to be loaded on Sorter shall have the following characteristics:

1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behavior of parcel cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

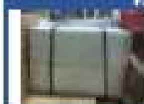
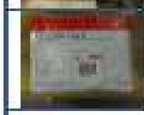
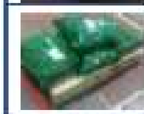
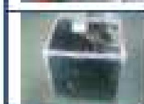
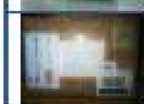
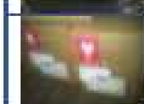
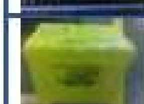
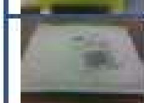
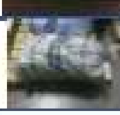
5.3 Parcel not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
2. Items that have a spherical or cylindrical shape.

3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

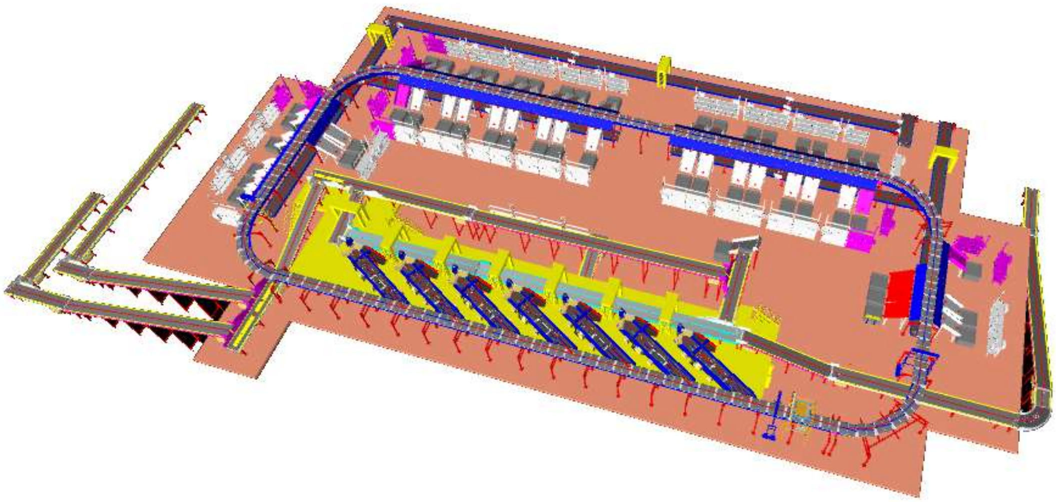
5.4 Pictures of Sortable Parcels

PARCEL TYPE		DESCRIPTION
	Strongly Strapped	A parcel banded with strapping material.
	Pre Manufactured Plastic Courier Pack <A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre Manufactured Plastic Courier Pack > A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed.
	Other Plastic Covered Item	A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item	A cardboard parcel coated in a packing paper.
	Cloth Covered Item	Often import parcels that are of irregular shape.
	Wine Box	A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box	A cardboard parcel is often the norm.
	Totes	A large plastic, lidded box.
	Flats	A parcel that often contains documents.
	Poly Bags	It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.

6. Proposed System Description

6.1 Objective

The purpose of this proposal is the design, manufacturing, installation, commissioning, testing and acceptance testing of a complete Sortation system to handle Parcels.



6.2 Summary of the System

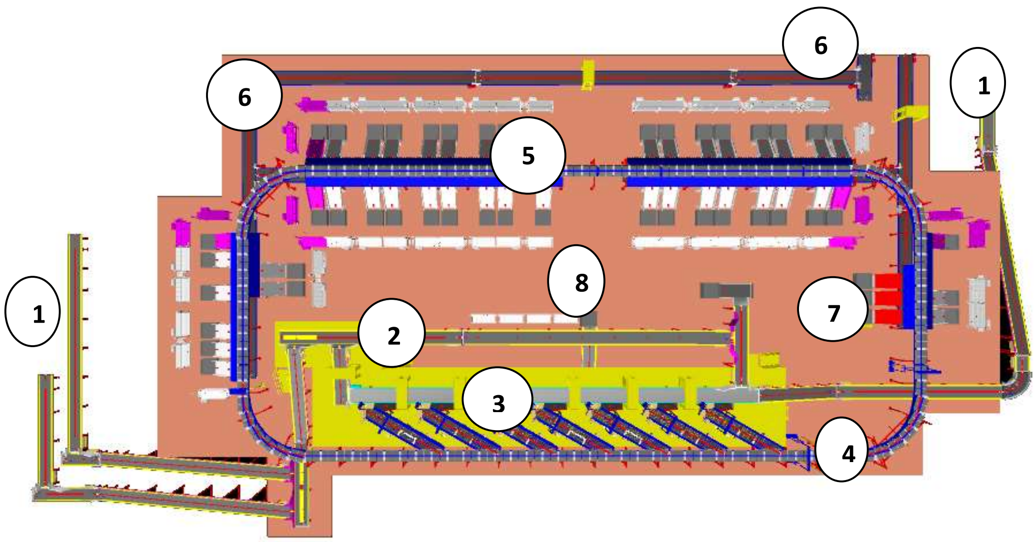
1. Bulk Infeed conveyors are used to transfer the Shipments from Ground level to Mezzanine Level to the VDS
2. Mixed Load of Parcels is dumped to the Induct Stations via loop VDS systems.
3. The VDS system feeds 7 Induct Lines.
4. Each Induct line is Equipped with a Loading Conveyor and online automatic weighing.
5. The operator picks up a parcel from the VDS Chute and places it on the Loading Conveyor with Barcode facing upwards.
6. These Induction Lines induct the parcels on the Sorter automatically onto to the Cross-Belt Sorter which then sorts the parcels into respective chute using data from Delivery's Sorting decision system. The Dimensions and image of the parcel is also captured on the Sorter.
7. The sorted shipments are discharged via sliding chutes into Bellow type collection bins.
8. Take out Conveyors are planned below the sorter. Once the Bags are filled, they are pushed onto the Take-out Conveyors after sealing and tagging.
9. Take-away conveyors terminate in spiral chutes (Delhivery Scope) that take the bags down to the ground floor.
10. Safety devices are considered as per Falcon Standards and referenced from previous projects.
11. Control and IT Systems to suit the requirement and as per Delhivery specification.

6.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput
2. Good Parcels conveying quality for the entire range of conveyable Parcels with a sorting conveyor speed up to 2.1 m/s.
3. Maximized sorting capacity throughout the entire sorter loop, including the two infeed zones.
4. Low occupancy of floor space in the building
5. Narrow discharge centers for increased number of splits in limited space
6. Scalable with increasing Business Volumes.
7. FALCON's CBS can adapt with changing business requirements by adjusting its speed, to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear

7. Layout View



Legend

1 – Infeed Line

2 – VDS Loop

3 – Feedlines

4– Main Loop

5 – PTL Chutes

6 – Takeaway Conveyors

7– Rejection Chutes

8 – non-conveyable sorting Station

8. Proposed System Capacity_ Calculations

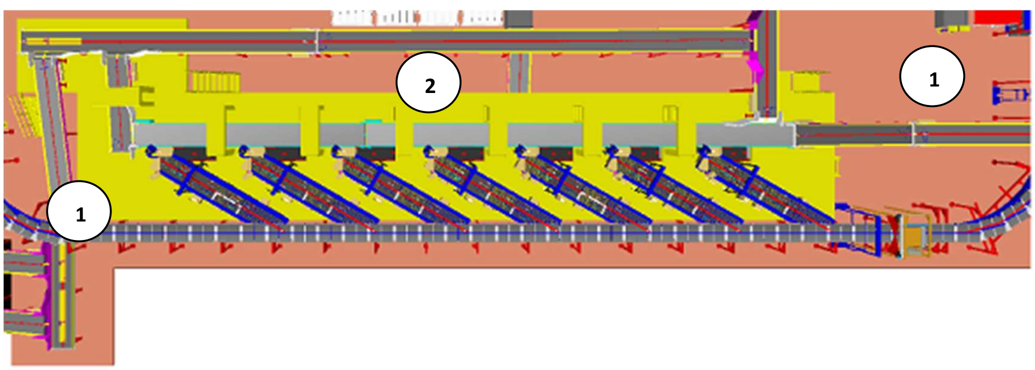
8.1 Sorter System Capacity

The following table shows the Throughput Calculation for the Sorter

Sorter Speed	2.1	m/s
Carrier Pitch (For dual belt)	1175	m
Carriers per Hour/ Zone	12600	CPH
Sorter Designed Throughput	12600	PPH
Rated TPH per Induct Line	2000	PPH
Number of Induct Lines	**7**	**QTY**

9. System description

9.1 Infeed System



Legend

1- Infeed Conveyor

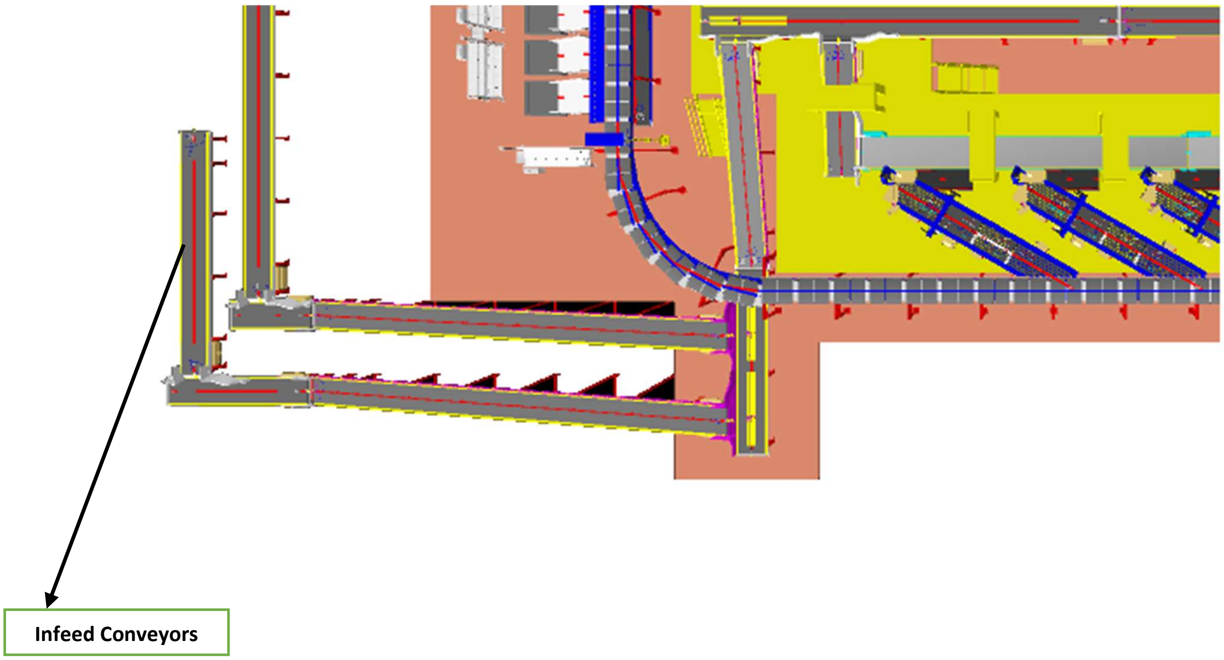
2- VDS Loop

9.1.1 Bulk Infeed Conveyor

Falcon's PVC Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

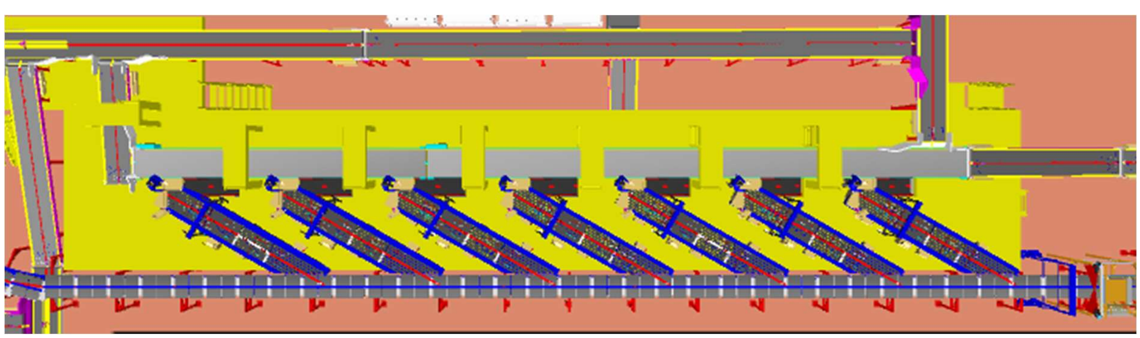
Some Silent Features of Falcon's Belt conveyor:

- Derco /Forbo Belts
- High Friction Belts for Inclines and Declines
- Side Guides



9.1.2 VDS Loop

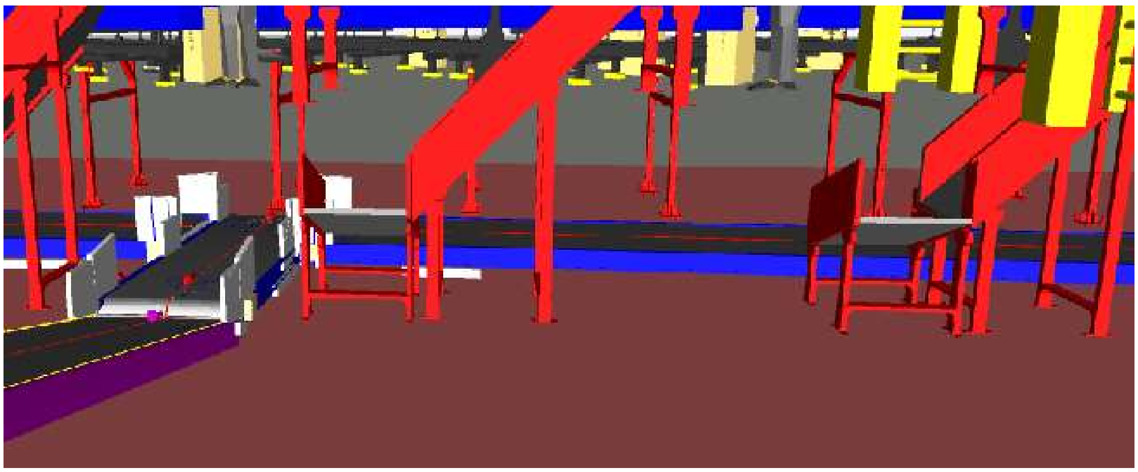
The loop Volume Distribution system enables seamless picking of incoming bulk flow of shipments from the infeed conveyor. The left-over shipments are re-circulated in the loop to repeat the process.



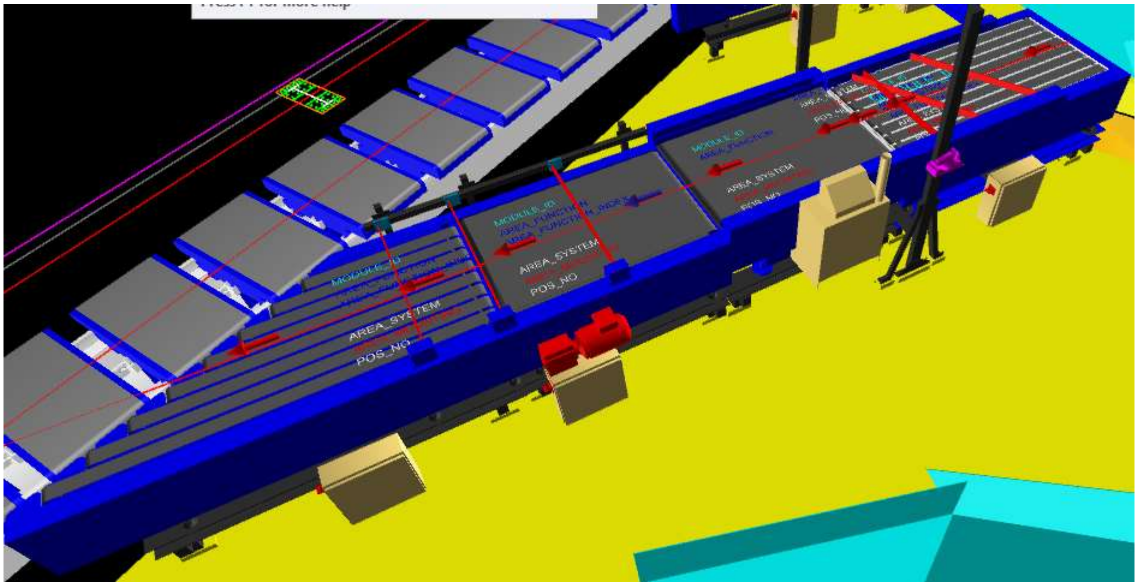
9.2 Irregular Shipments take out Conveyor

Provision of Chutes are provided at Induct Stations to Take out Odd Shaped Shipments from the System.
Operator identifies these shipments visually or by Size Gauge and drop them onto the chute and Shipments slides through the Chutes on the Conveyor.

The conveyor terminates at the non-con manual sorting station where they are manually sorted into PTL racks manually.



9.3 Feedlines



1. The Induct zone is equipped with 7 Induct lines.

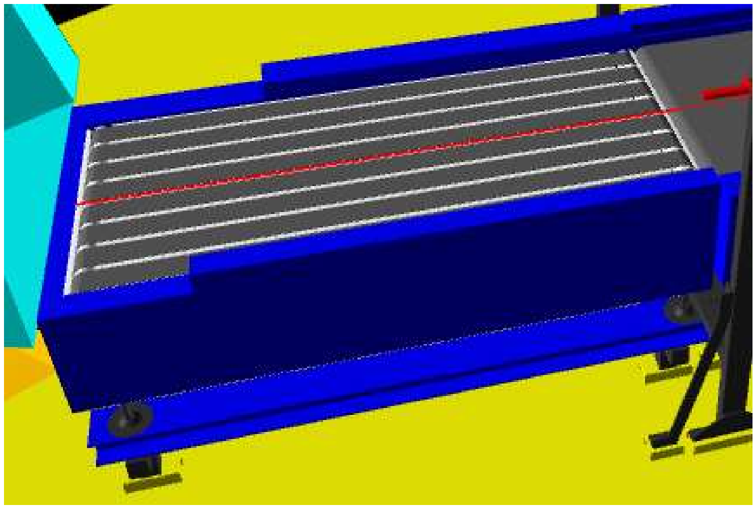
2. Each Feedline has four Conveyor Modules- 1. Loading & Orientation Conveyor, 1 weighing conveyor, 1 Intelligent feeding unit conveyor and 1 angle merge conveyor

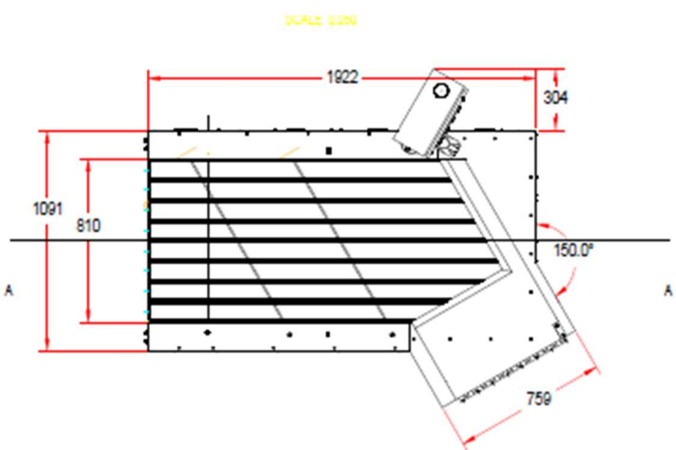
3. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Parcel on the conveyor and prepare the Cross-Belt Cell for receiving the parcel with high precision.

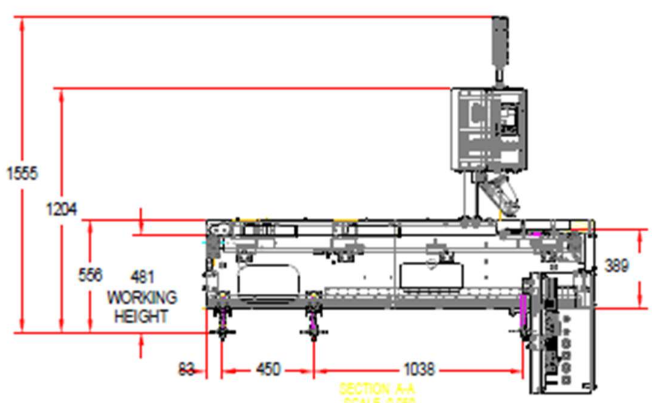
4. Over the feedline, parcels are evenly spaced and smoothly transferred to the main Loop.

9.3.1 Loading & Orientation conveyor

Parcels are loaded on this conveyor with proper orientation so that the barcode sticker is at the top. Orientation is achieved with the help of fixed aligner.





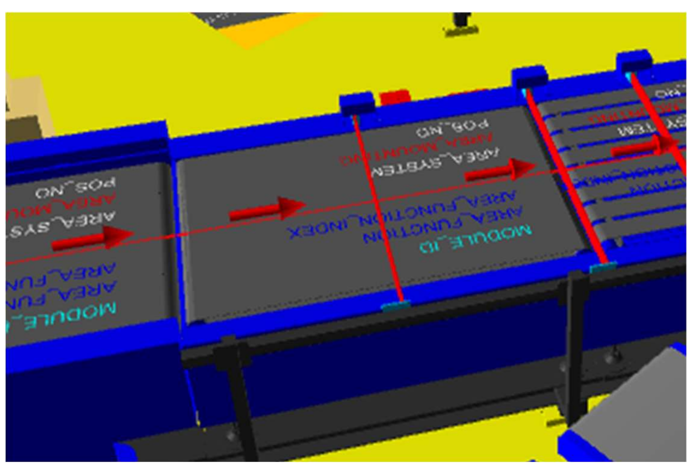


9.3.2 Weighing conveyor

This conveyor automatically measures the weight of the parcel.

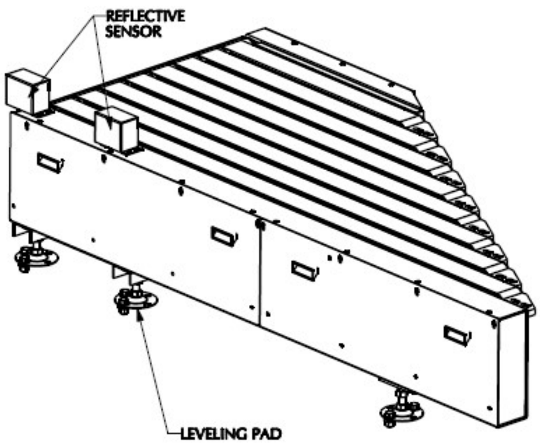
9.3.3 Intelligent Feeding Unit

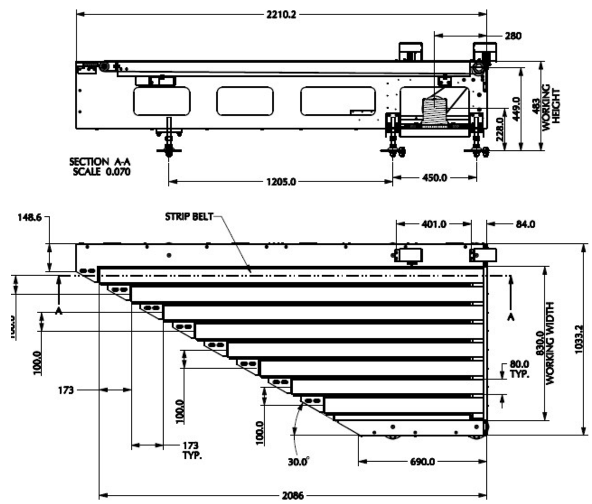
This conveyor is a variable speed special purpose module that creates space between parcels as well as regulates feeding on the intelligent merge.



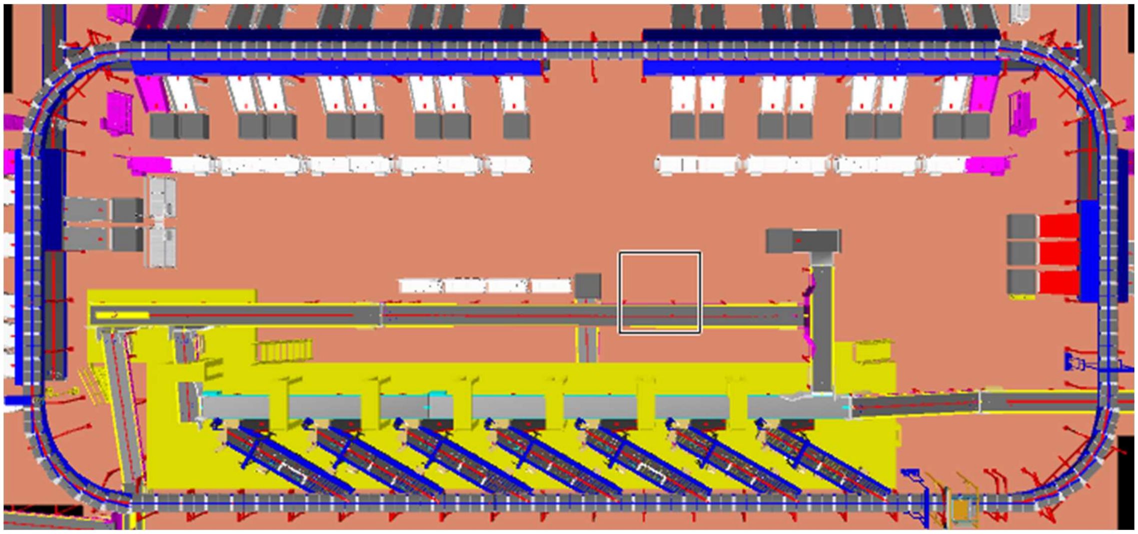
9.3.4 Intelligent Merge Conveyor

This is 30° triangular Dual Belt high-speed conveyor used for inducting parcel/boxes directly on to the sorter. The Belts are Strip Belts for smooth parcel movement.





9.4 Main Loop



Main Loop

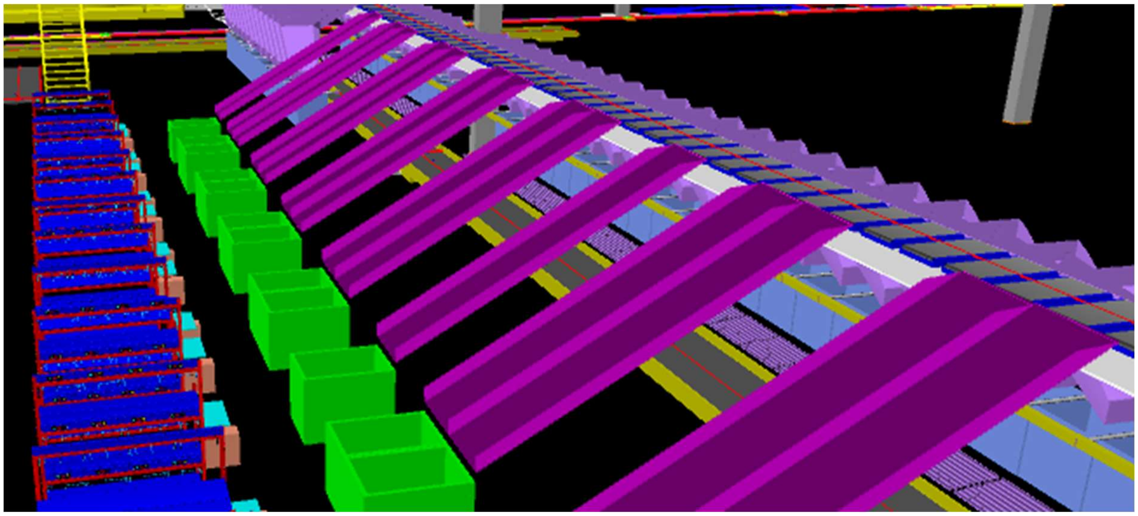
1. The proposed Sorter will be installed on the mezzanine (4000 mm clear height from ground). The Sorter will run at 2600 mm w.r.t. the mezzanine level at 2.1 m/s speed.

2. We have provided Dual Belt Carriers with 1175 mm pitch and Belt Size of 800x495 mm. Please note in dual belt carriers each individual belt can carry one parcel provided the parcel dimensions are within the single belt carriage's permissible limit. Please note. dual belt features, i.e., the ability of the sorter to place over-sized products on two belts, has not been considered due to shortage of space and lack of utility for the parcel profile which Delhivery intends to run on the sorter.

3. As Parcels pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the shipment reaches respective chute, carrier carrying the shipment for the chute, will actuate and drop the shipment in the assigned chute.

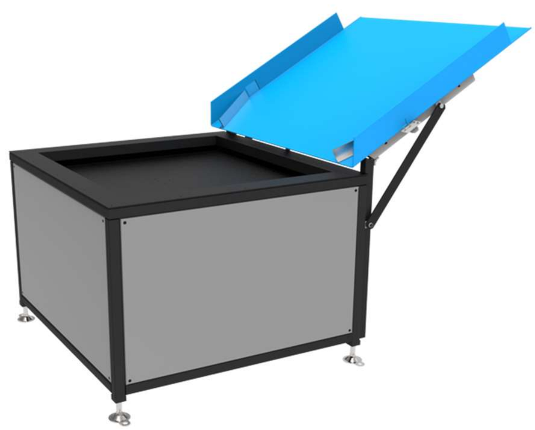
9.5 Output Chutes

Falcon designed Sliding type chutes



1. **PTL Chutes with spring loaded bins**

   These chutes drop the parcel in the bellow type spring loaded bins for secondary sorting by means of PTL racks



Chute IOs	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation.
Tower Lamp	To indicate the operator action on a chute

9.6 PTL Racks & Modules

PTL Racks with PTL light modules has been provided for secondary sorting of parcels that have been dropped in the PTL chutes.

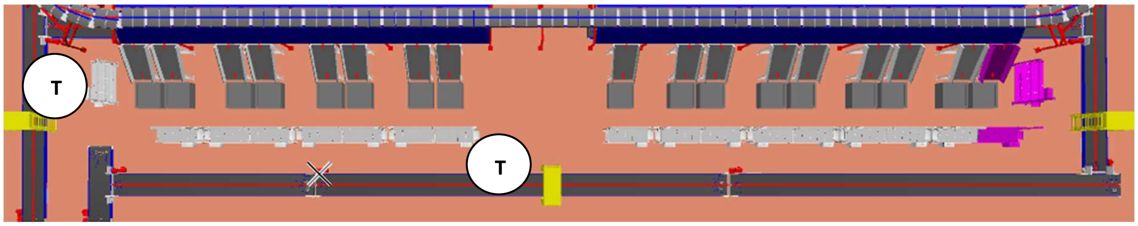
Rack Configuration: 3 Rows, 4 Columns, total of 12 modules/rack in one chute. There are 5 no's dual PTL chutes where there are 2 racks of total 24 PTL modules are mapped to a single chute.

- Operator will pick up a parcel from the PTL chute bin and scan the parcel barcode by handheld scanner
- The light of the particular block in the PTL rack which has been mapped to the destination location of the parcel will glow.
- Operator will place the parcel in the bag framed in that module and acknowledge by pressing the confirmation button on the module.
- This process will be repeated for all parcels falling in that chute bin.
- Once a bag is full, operator will take out the bag paste specific tag on bag seal and place it on the bag take-away conveyor.
- New bag will be placed in the empty module .
- Due to space constraint control boxes will be placed at the back side of the racks.
- However an extended plate of not less than 205 mm width will be provided at the side of the racks for placing bar code label printer(printers in Delhivery scope)

10.7 **Bag Take Out Conveyors**

For Direct Bagging Chutes, Conveyor Network is planned to Take out Order bags from the System.

Once the Bags are filled, they are pushed onto the Take-out Conveyors



T **Take Out Conveyors**

10.Proposed System Technical Details

10.1 Mechanical Equipment- Bill of Materials

Pos.	Qty.	Description	Value
FM1	**1**	**Conveyor Package** • Powered Belt Conveyors • Powered Turn Conveyors (90 Degrees) • Modular Conveyors	~344 Metres (33Conveyors) 1 Nos. ~30 Metres (2 Nos). .
FM2	**7**	**Feedlines** Consists of • Single Operator Loading Conveyor. • Weighing System • Dynamic Spacing System • Angle merge • Hand Scanners for Manual Scanning	1 Set 1 Set 1 Set 1 Set 1 No.
FM3	**1**	**Sorter** **1 Loop Cross Belt Sorter** • Sorter Height • Sorter Length • Sorter Speed • Sorter Drive • Carrier Pitch Including: • Standard Sorter Supports • Empty Carrier Detection System • Product Centring System • Camera based Barcode reader • Dimension Scanning System • E-Stops • Hooters • Fencing • Netting • Overshoot Assembly	2600 mm(from mezannine) ~135 m ~2.1 m/s Linear Motor 1175 mm
FM4	**1**	**Chutes** • PTL Chutes • Rejection Chutes • Irregular Chutes • Chute Full Sensors • Tower Lights • Contingency Chute • Spring Loaded Fixed Bin	45 Nos 3 Nos 7 Nos 48 Nos 48 Nos 1 Nos 50 Nos

FM5	**1**	**PTL** • PTL Racks • PTL Modules • PTL Control Box • PTL Barcode Hand scanner 1-D	648 Nos 600 Nos 50 Nos 45 Nos
FM6	**1**	**Steel Works** • Operator and Maintenance Platform **Please note- The current Proposal does not include the Price of Steel Works and the same shall be discussed separately with yourself.**	~2304 SQM

10.2 Electrical Equipment

Electricals			
FE1	**1**	**Consists of** • Main Power Distribution panels • Main Control Panels • Feedline Control Panels • Conveyor Drive Panels • Sorter Drive Panels • Network Switches • Field Cabling • Cable trays • Remote I/O Panels	Included

10.3 Control System

Components			
FC1	**1**	**Controls and IT Infrastructure Controls** • Weigh Scale Control • Barcode Scanner Controls • Dimension Scanner Controls • Conveyor Package Controls • Sorter Control • Induct Control • Smart Chutes Control **IT** • Control IT Software Package • High Availability Server Setup (Microsoft Clustering based) • Networking Switches	Included

11.Description of Components of Equipment

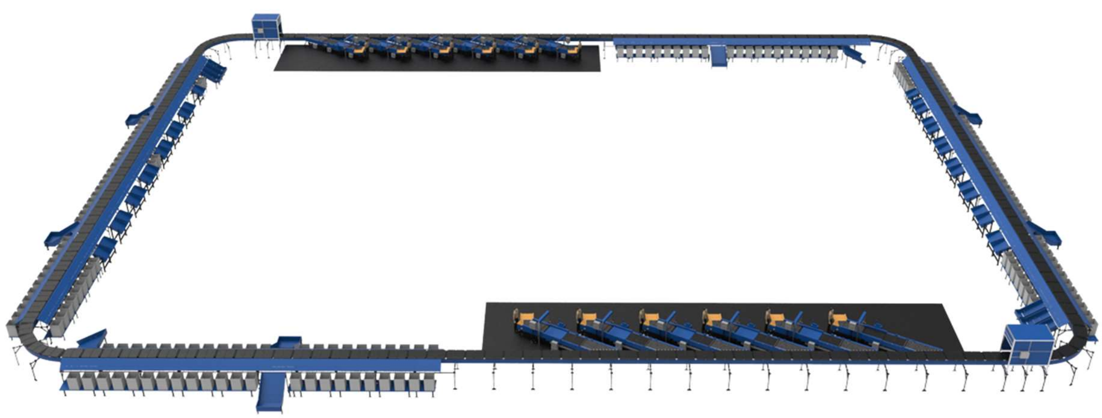
Elements of the sorting system

1. Cross Belt Sorter
2. Scanning & Sensing on Sorter
3. Conveyor System
4. Steel Works- Mezzanine & Staircases

11.1 Cross Belt Sorter -

Cross Belt Sorter is capable of sorting extremely high volume of versatile products in a gentle manner.

Falcon's Cross belt sorter is powered by high efficiency linear motors and is based on 100% non-touch actuation technology leading throughput capabilities with extremely low noise levels.

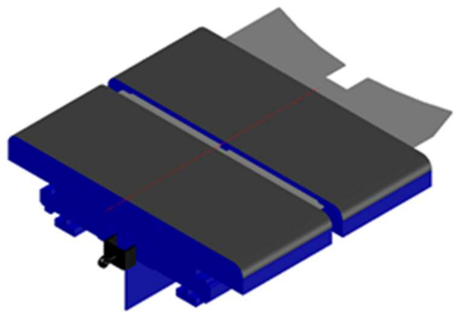


Falcon's Cross belt sorter is modular in design. It can be easily extendable as per future requirements

11.1.1 CBS Carrier

Falcon Autotech' s CBS offers one of the highest Belt widths to Carrier Pitch Ratio in the market today. This additional belt width makes the system capable of handling larger product sizes without compromising on throughput. This also means reduced "dead area" between Carrier belts which drastically reduces the amount of "inbetweeners "and non-sortable parcel recirculation.

We have provided CB dual-belt Carriers with 1175 mm pitch. Parcel sizes up to 500x450 mm will go on a single carrier. Please note we have not provided dual carrier features in this project due to lack of space and lack of utility for the parcel profile being handled by Delhivery.



11.1.2 Servo Roller

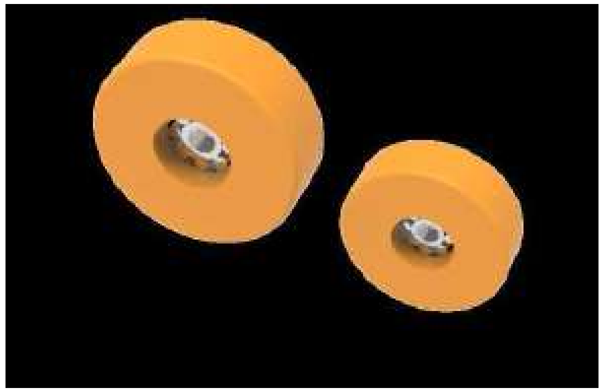
High Powered DC Drive Servo Roller are used to actuate the carrier belts, thereby eliminating the need for complicated drive transfer mechanism, and simplifying the system installation and maintenance.

11.1.3 Chassis

Falcon Autotech's Cross Belt Carrier Chassis is made up of lightweight Aluminum which makes them light yet sturdy. This reduced weight leads to substantial "Power Savings" over a considerable period of usage.

11.1.4 Carrier Wheels

Carrier wheels that are thoroughly tested and proved for long life cycle



11.1.5 Non-contact based linear motor drive

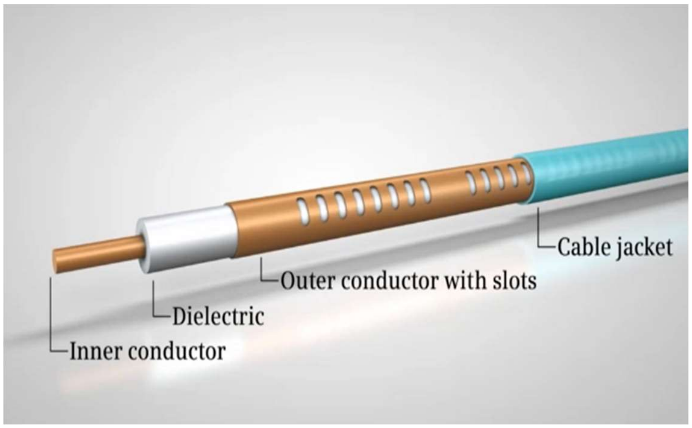
Extremely **High Energy Efficient** Linear Induction Motors that can be configured at variable speeds depending upon the operational requirements giving you the maximum flexibility when needed

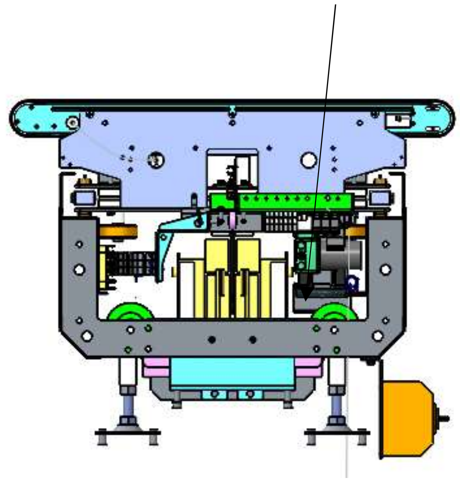
11.1.6 Power Transmission

Power transmission to Carriers over sliding contacts that require low maintenance and offer high levels of reliability.

11.1.7 Data Transmission

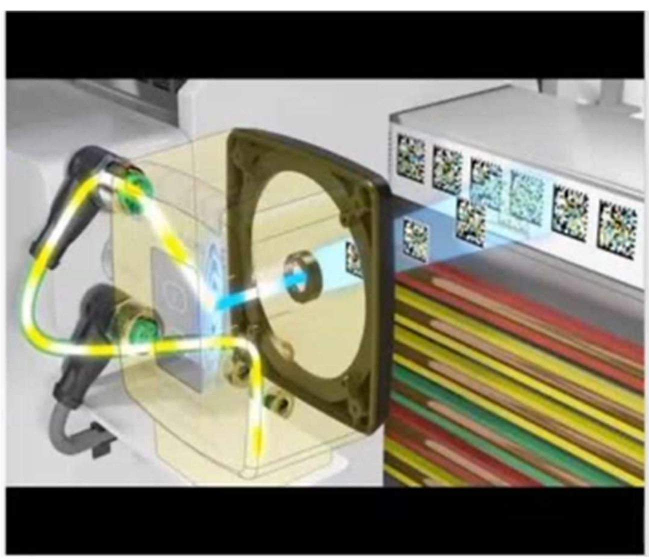
The R-Coax cable is used for data distribution in the sorter. This is a leaky wave cable, that runs throughout the sorter length transmitting the data. There is one antenna present in the suparmaster carriage, which keeps on receiving the signal from this cable while on the move, wirelessly.





11.1.8 Carriers positioning System

Positioning system determines the exact location of each carrier in the loop at any given point in time. There is a plastic tape strip of barcodes (QR Codes) that is running along the sorter loop, which is continuously scanned by scanners placed on a Carrier (Master Carrier)



11.1.9 Technical specifications of Sorter

Items	Make
Sorter Carrier Type	CBS Dual Belt
Sorter Speed in m/s	Up to 2.1 m/s
Sorter Loop Length	~135 m
Length of the Sorter	Layout Attached
Width of the Sorter including Chutes	Layout Attached
Sorter Actuation Technology	Electric
Noise Level	<75 Db
Chutes	
Chutes + Rejection	45+3
Height of Sorter	~2600 mm w.r.t. Mez
Track Material & Type of Design to move the Carrier trolleys	~Steel Track
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch	1175
No. of Carriers	115
Drives	
Motor/Drive type	Linear Induction Motor
Power Consumption	Will be specified during DAP
Induction Mechanism to the Sorter	
Automatic Feedlines	7
No. of Barcode Scanners	1
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Parcel Positioning System	1
Parcel Centering System	1
Empty Carrier Detection System (Camera Based)	1
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Layout Attached

11.2 Scanning & Sensing on Main Sorter Loop

11.2.1 Automatic Barcode Scanning & Dimensioning System

Top Sided Barcode Scanner to scan parcel 1D codes and 2D codes. The Scanner is also capable of Archiving Shipment Images in Real time.

VMS 4200 – Installed in this tunnel to capture dimensions of the parcel.

Handheld Barcode Scanners are provided to run the machine on Semi- Automatic Mode, if required

11.2.2 Parcel Positioning System

Falcon Cross Belt Sorter is equipped with Multiple parcel positioning Scan Tunnels. A tunnel comprises of a series of high Precision Laser Scanners that detect the absolute position of the Parcel on the Cross-Belt Carrier and prepares the Carrier for discharging the product accurately into the chute (virtual centring) and physically bring the product to the centre of cross belt carrier (Physical centring).

11.2.3 Empty Carrier Detection System

This system detects the empty carriers on the sorter and prepares the Feedlines to load the parcels on the empty carriers. (Camera based)

11.3 Belt Conveyor

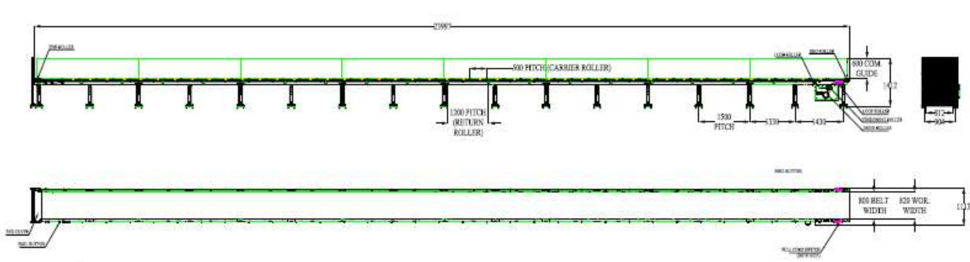
This paragraph presents the main characteristics concerning straight or inclined conveyors.

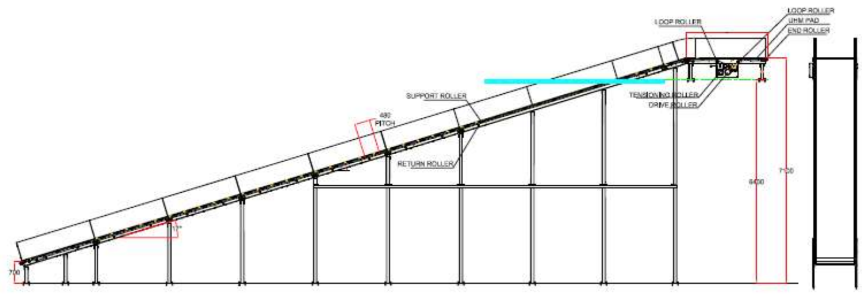
Falcon's Belt Conveyors are modular & robust in design, used for smooth conveying of products over Straight, inclined and declined paths. MS profile is used to build conveyor frame.

The conveyors are supplied with the necessary supports and bolts to fix them to the supporting plane, as well as with the junction elements allowing easy and jam-free passage from one conveyor to the other.

Some Silent Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.





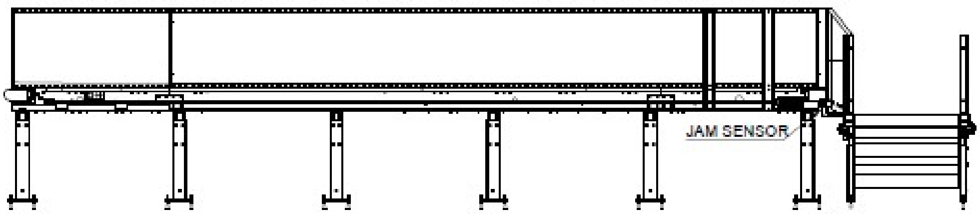
PVC Belt Horizontal & Inclined Conveyors

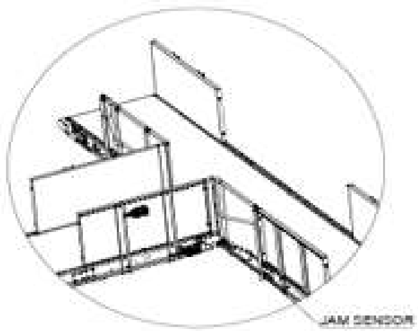
11.4 90-degree Curves (Ref Picture)

The design of the curves will be robust and easily maintainable. The metal frames of the belts are not deformable to prevent belt misalignment. The belt guide assembly will include removable parts to allow quick replacement in case of damage.

11.5 Waterfalls

A 100 mm of drop has been provided at 90-degree transfers.



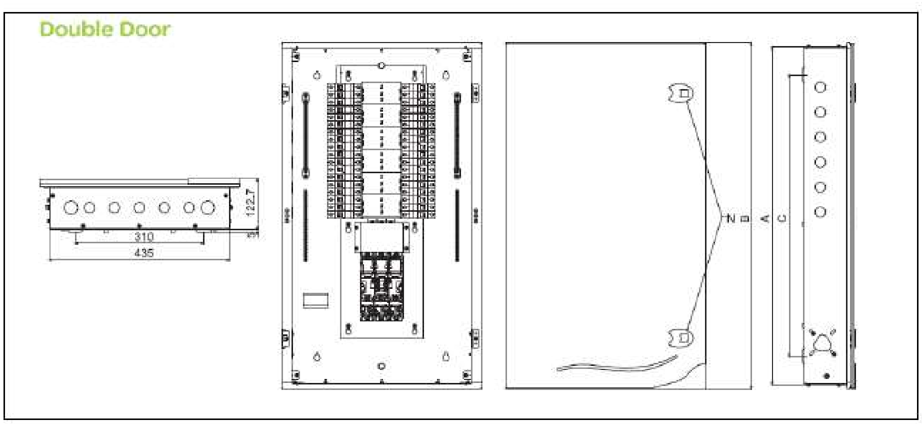


12. Electrical System

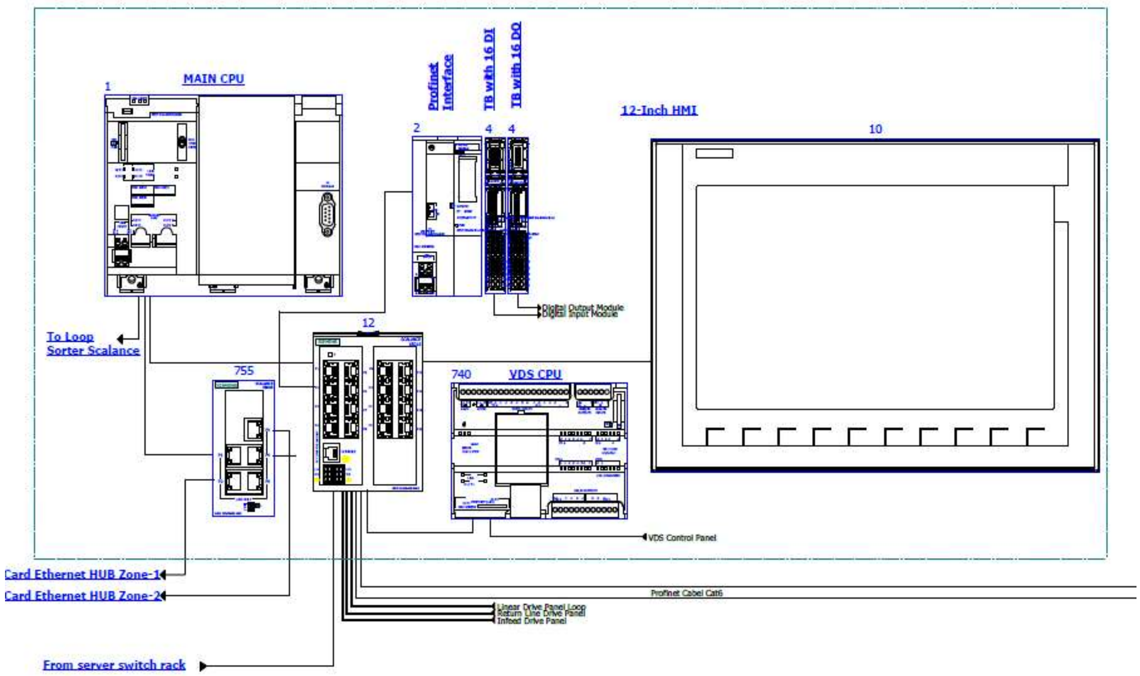
Main power supply will supply Falcon's PDP (Power Distribution Panels) electrical cabinets.
PDP cabinets supply the entire system via secondary cabinets:

- Main Control Cabinet
- Induct Control Panels
- Remote Cabinets for Sorter I/O
- Scanner Control cabinets

12.1.1 Reference Picture of Power Distribution Panel



Main Control Panel (Reference)



Engines

Three-phase alternating current motors (Induction) will be used through a frequency converter.
The engines will be coupled with a converter to improve consumption and reduce the carbon footprint.
All motors will have appropriate IP ratings

Sensors

The sensors will be supplied, standardized by type, with connector, with a cable length suitable for easy extraction, suitably protected from possible impacts.

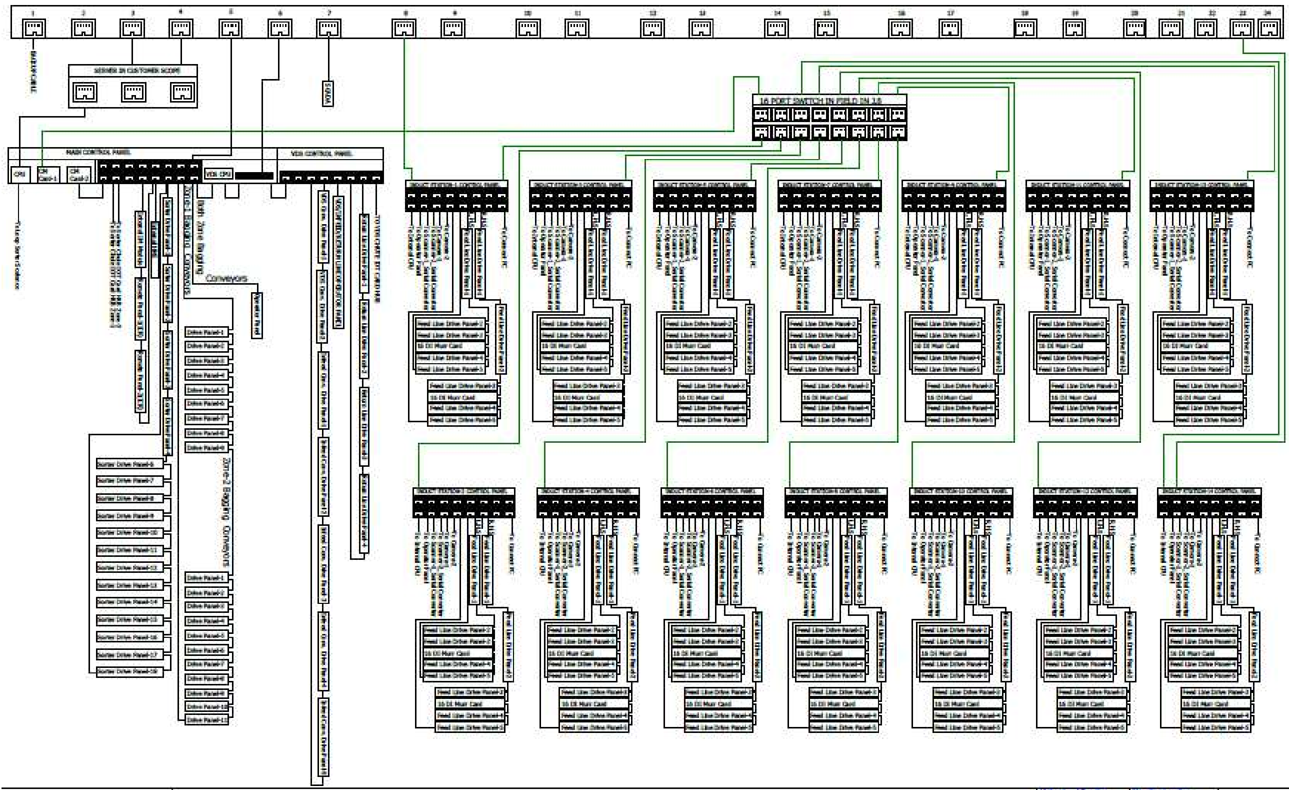
Control command

The proposed solution is based on SIEMENS Programmable Logic Controller technology (PLC) platform. The entire system will be logically divided into zones (Sorter/ Feed Line/ Loop), each managed by a PLC. The planned primary communication protocol is going to be ProfiNet.

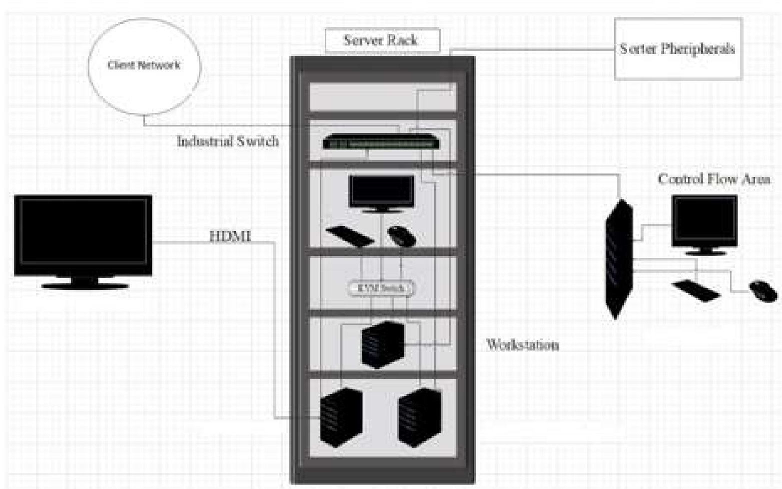
Conveyor interface

The frequency converter of each conveyor allows the acquisition of the signals of the sensors/actuators/GIOs associated with it (e.g. conveyor end detection photocells, blockage detection photocells). Each frequency converter will be connected in series by means of the ProfiNet field bus

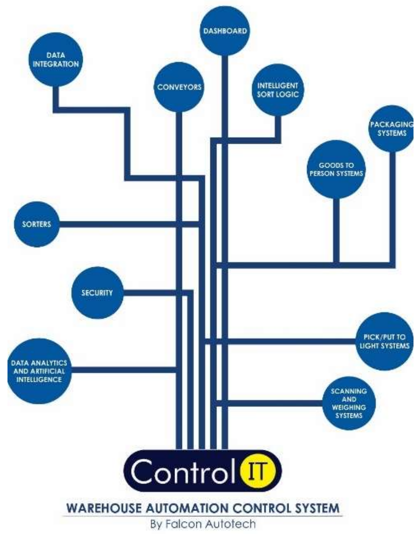
13.Controls Architecture (ref.)



14. IT Architecture (Ref.)



15.Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

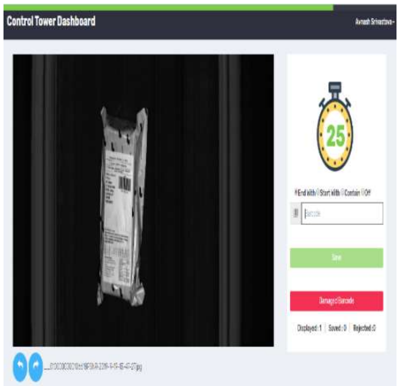
API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

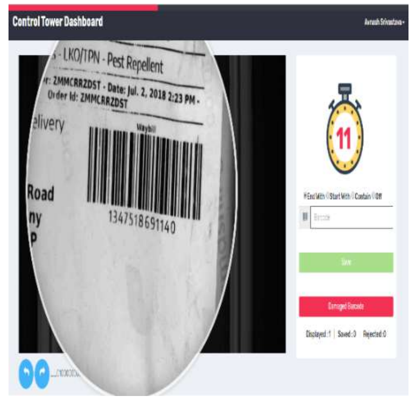
16. Key Components Make

Items	Make
Belts	Forbo/Derco
Rollers	Falcon
Cross Belt Carriers	Falcon
Linear Induction Motors	Falcon
Feed Line Motors	Induction Motors – SEW/ Lenze
Barcode Readers	SICK / Cognex
Wireless Barcode Scanner	Honeywell/Motorola
Load Cells	MT/Bizerba
Volume Scanners	SICK
Encoders	SICK
Sensors	Sick/Leuze/ P&F
PLC	Siemens’s
Control Panels	Rittal/ BCH
MDR Rollers	Falcon/ Pulse/ Interrol
Power Transmission Systems	Vahle

17. USPs of Falcon's Cross Belt Sorter

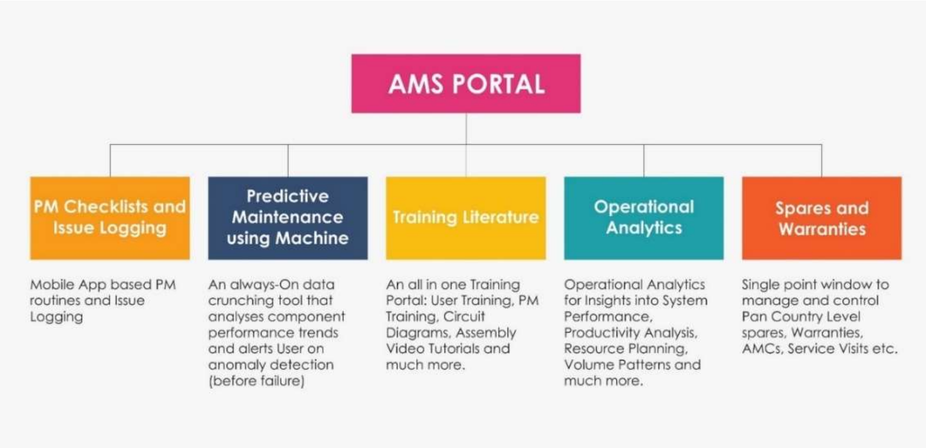
Real Time Video Coding – this new innovative feature of Control IT software directly focusses on reducing the rejections to approximately zero which arises out of barcode unreadability while the sorter is running. This allows a remote-Control Tower to Decode images from a PC and relay the decoded value back to the sorter in real time and reduce Barcode read rate related rejections drastically. This is a flagship feature in Falcon's Cross Belt Sorters.



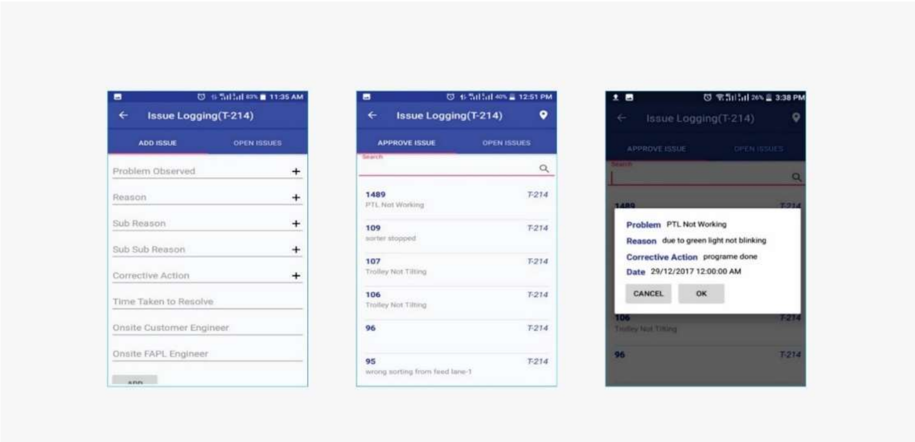


- Falcon Autotech' s Cross Belt Carrier chassis is made up of lightweight Aluminium which makes them light yet sturdy. This reduced weight leads to Potential "Power Savings" over a considerable period of usage.
- Carrier wheels with large diameter are used which reduces number of revolutions for the same Track Length, thereby reducing wear and tear
- CBS Carriers can discharge a parcel anywhere along the track of the sorter, thereby ensuring 100% flexibility in Chute positioning even after the system installation. This also means that sorter can discharge products on turns, thereby increasing the chute density and reducing the overall system footprint as well as cost'
- Variable Electric Discharge Acceleration depending upon the product profile.
- Contact based Linear Motor Drive options for-
    1.) Low noise levels (<75 Db) meaning a healthy work environment for the work force.
    2.) Reduced mechanical break downs leading to high System overall availability (>98.5%)
    3.) Extremely low maintenance requirements due to no wear element
    4.) Back-up Drive units for In-built system redundancy

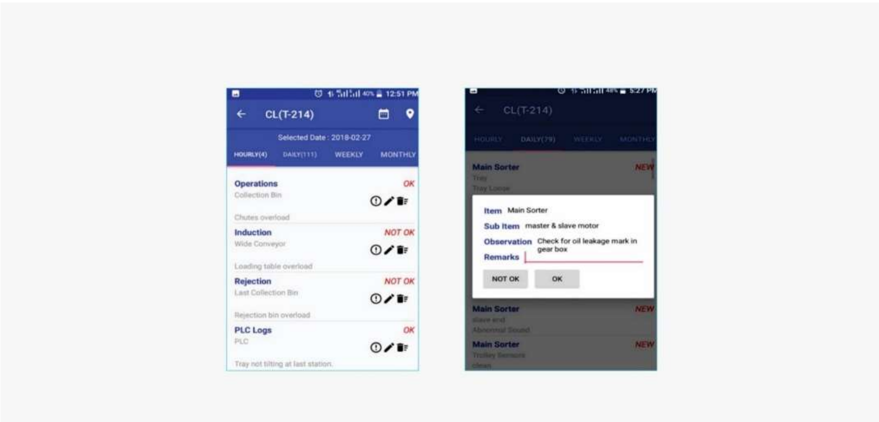
18.Automation Management System (AMS)



a. Issue Logging

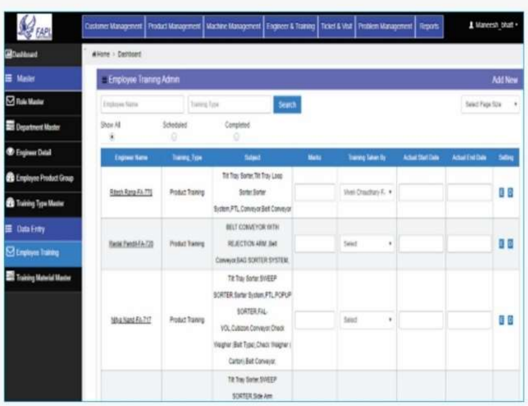


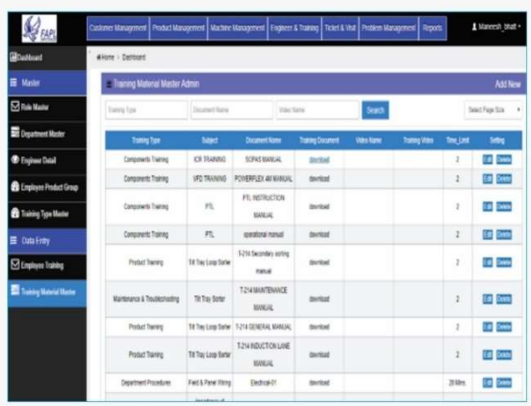
b. Preventive Maintenance Check List



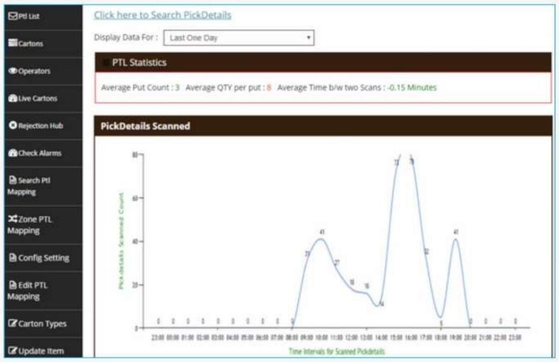
c. Predictive Maintenance

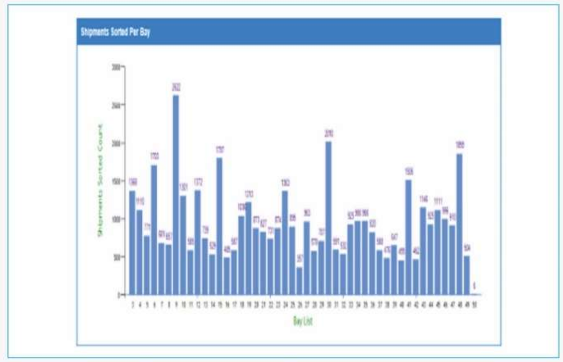
d. Training Portal





e. Operational Analysis





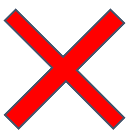
19.SOPs for feeding

i. Initial Checks

1. Clean dust from sensors mounted on all conveyors and sorter (ensure that sensor should not be blinking or permanent on)
2. Ensure operators are at their designated stations only
3. Start machine by pressing START (green) button

ii. Parcel Selection

1. Parcel size:
   Minimum Parcel Size Should be 100(L) x 10 (W) x 5(H) mm
   Maximum Parcel Size should be 500(L) x 500 (W) x 400(H) mm
2. Parcel weight: Max 7 KGs
3. Parcels with irregular base (unstable) should NOT be fed on to the sorter
4. Parcels with barcodes on the highest surface (max of L, B, H) should NOT be fed in automatic mode, as they tend to topple

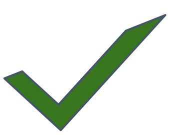


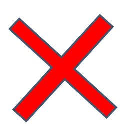


iii. Feeding

1. Parcel should be placed on a flat base with barcode facing on the top side

2. For poly packs / flyers, ensure the loose section of the poly is tightly wrapped around the solid section of the parcel to avoid double trigger to the sensor (Double trigger to sensor results in lesser throughput and false alarms)






20. Warranty Period

Falcons offered System comes with a standard warranty of 12 months from successful commissioning of the system or 15 months from the date of completion of delivery, whichever is earlier. In case of delay in lifting the material by client, the 15 months warranty period will start on the 15th day after material readiness intimation is formally communicated to client

Post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
    - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
    - Negligence or abnormal use of equipment.
    - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
    - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

21. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- **Air Compressor**
- **Construction Power**
- **Steel Works- All steel works including the sorter mezzanine as well as operator platforms at the induct zones and maintenance platforms for o/h conveyors will be in Delhivery scope.**
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.

22.Commercial Offer & Terms and Conditions

22.1.1 Itemised Cost of Sorting System

Heads	Sub heads	Qty	Cost
Conveyors	Powered Belt conveyor	204	₹ 1,00,10,825
	Turns	1	₹ 7,70,000
	Modular conveyor	30	₹ 23,63,533
Sorter	Inducts	7	₹ 77,61,635
	Loop	135	₹ 3,75,61,180
	Cognex Scanning & Dimensioning System	1	
		1	
Weighying conveyor		7	₹ 55,00,000
Bar code scanner for semi Auto		7	₹ 1,26,000
Chutes	Sliding Chutes+sensor+I/O+Tower Lamps	56	₹ 25,05,427
	Spring loaded Bins	50	₹ 19,25,000
PTL System	Hand held scanners	45	₹ 68,67,920
	Modules	648	
	Frames	600	
	Control Boxes	50	
	Rejection Frames	1	
Bag Conveyors		140	₹ 63,67,126
Misc	Tool Kit	1	₹ 79,800
Controls & IT		1	₹ 30,11,223
Project Management		1	₹ 4,53,876
Packing & Forwarding		1	₹ 16,92,993
Installation & Comissioning		1	₹ 42,32,483
Total			**₹ 9,12,29,020**
BCA			**4.5%**
Nett Price			₹ 8,71,23,715
Old Discount			1 %
Nett Price			₹ 8,62,52,477
Price Difference(ICR to Cognex)			₹ 4,06,295
Final Price			**₹ 8,58,46,182**

22.1.2 Price Terms

The Total Price is to be understood.

- Ex Works
- Taxes: Extra as applicable
- Excluding Freight & Unloading & MHE's as required.
- Including Server PC
- Including Packing as described
- Excluding Steelwork i.e. (Mezannine , Operator Platform and Stairs)
- Including Installation Supervision Cost & Commissioning.
- Scaffoldings, Man Lift etc. required for reaching the maintenance platforms shall be in customer scope.

*Material Unloading & Laydown area is in customer scope.

22.1.3 Payment Schedule

The Supplier Quotation is based on the following payment schedule:

Payment %age	Project Stage
40%	Advance along with PO
50%	With 100% taxes within 30 days of material reaching at Site
10%	After Installation & Commissioning

(*): *All invoices (if dues as per payment terms) will be cleared with in 30days from the date of receipt of undisputed invoice by the buyer.*

22.1.4 Schedule for Delivery and Installation

The complete system is to be delivered, installed, and commissioned by 10th Sep 2023.

* *Detailed Project Plan will be shared during DAP.*